CHAPTER-1

ROW ECHELON FORM

1. The process of converting a matrix to Row Echelon Form is called:

A) Matrix inversion

B) Gaussian Elimination

C) LU Decomposition

D) Gram-Schmidt Process

2. In Row Echelon Form, what must be true about rows consisting entirely of zeros?

A) They must appear at the top

B) They must appear in the middle

C) They must appear at the bottom

D) They cannot exist

3. Which elementary row operation is NOT valid when reducing a matrix?

A) Swapping two rows

B) Multiplying a row by a non-zero scalar

C) Adding a multiple of one row to another

D) Multiplying a row by zero

4. What is the pivot in a row?

A) The last non-zero entry in a row

B) The first non-zero entry in a row

C) The largest entry in a row

D) The entry on the main diagonal

5. How many pivots does a 3×3 identity matrix have in Row Echelon Form?

A) 1

B) 2

C) 3

D) 0

6. A matrix is said to be in Row Echelon Form if:

A) All pivots equal 1

B) All entries below each pivot are zero

C) All entries above and below each pivot are zero

D) The matrix is square

7. How many pivots does a 3×3 identity matrix have in Row Echelon Form?

A) 1

B) 2

C) 3

D) 0

8. Which statement about Row Echelon Form is FALSE?

A) Every matrix has a unique Row Echelon Form

B) Every matrix has a unique Reduced Row Echelon Form

C) Row Echelon Form is not unique but Reduced Row Echelon Form is

D) Row operations are reversible

9. Which of the following is a requirement for a matrix to be in Row Echelon Form ?

A) All entries must be 1 or 0

B) The pivot in each row must be to the left of the leading entry in the row below

C) The pivot in each row must be to the right of the leading entry in the row below

D) All diagonal entries must be 1

10. What happens to the solution set when you multiply a row by a non-zero scalar?

A) It remains unchanged

B) It doubles

C) It becomes empty

D) A new solution is added

ANSWERS:

QUESTION NO	ANSWER OPTIONS
1)	B)
2)	C)
3)	D)
4)	B)
5)	C)
6)	B)
7)	C)
8)	A)
9)	C)
10)	A)